Team Number: TEAM_NO

M1 — Automatic Category Suggestions for Wikimedia Commons

Problem Statement 3 · Approach document

1. Problem understanding

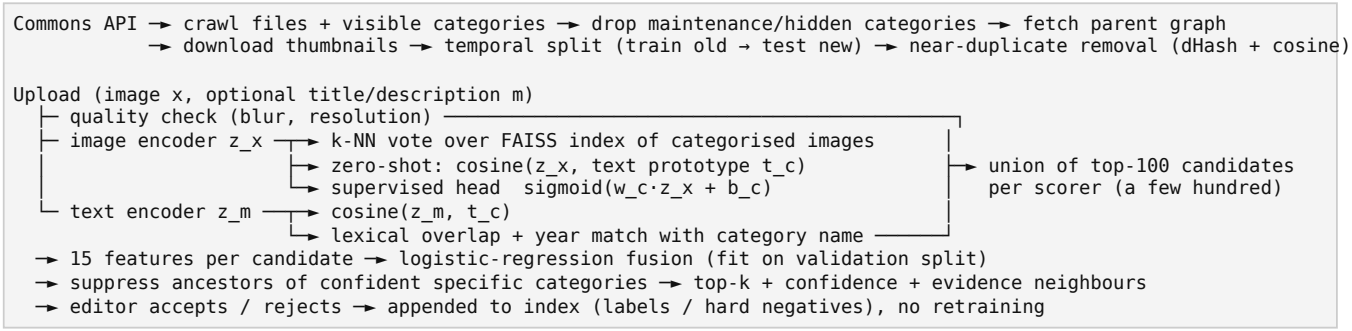
Given a newly uploaded image x and optional metadata m (title, description, date), return a ranked list of the top-k Commons categories for an editor to review. This is **extreme multi-label classification**: millions of categories, organised as a (mostly acyclic) graph, with a heavy-tailed frequency distribution. Key properties we design for:

- **Heterogeneous labels:** objects and scenes are visual; events and abstract concepts need text and metadata (dates, places, captions).
- **Noisy, incomplete labels:** editors file under the most specific category, so ancestors are correct but unlabelled — a missing label is *unknown*, not negative. Maintenance categories (licences, "Uploaded with…", contest and camera categories) describe process, not content, and must be removed.
- **Open label space:** categories are added daily; the system must accept them without retraining.
- **Operational needs:** generalise to *new* uploads, handle varied quality and resolution, scale to daily upload volume, and be open and explainable, in line with Wikimedia principles.

2. Idea / abstract

We treat categories as **retrievable objects** rather than a fixed softmax output. A frozen vision–language encoder (CLIP/SigLIP) maps images and text into one embedding space. Four complementary scorers each propose candidates: (1) **k-NN voting** over already-categorised images, which also provides the explanation ("these similar files are in this category"); (2) **zero-shot text prototypes** of category names, covering rare and brand-new categories; (3) a **supervised multi-label head** initialised from those text prototypes and trained with asymmetric loss to tolerate missing labels; (4) a **metadata scorer** (title/description embedding, word overlap with the category name, year agreement) for events and places. A **calibrated stacker** fuses the scores on a small candidate set, and a **hierarchy post-processor** prefers the most specific category. New images and categories are added to the indexes immediately; editor accept/reject decisions feed back as labels and hard negatives.

3. High-level pipeline



Requirement	Design response
Objects, scenes, events, abstract	Visual scorers for objects/scenes; metadata, year matching and text prototypes for events and concepts.
Accuracy on new images	Temporal split; dedup across splits; fusion weights fitted only on a held-out validation period.
Long tail / new categories	Zero-shot and lexical scorers need no examples; text-initialised head keeps prior knowledge for rare classes.
Missing labels	Asymmetric loss with margin; unlabelled ancestors of true labels excluded from fusion training (treated as unknown).
Efficient & scalable	One encoder pass + ANN lookup per image; FAISS flat / HNSW / IVF-PQ; fusion only over candidates; asynchronous, never blocks the upload.
Varied quality	Blur/resolution flags as fusion features; fusion trained on degraded copies (blur, JPEG, downscale) and on copies without metadata.
Keeps learning	Index append for new images/categories; feedback log of accept/reject events for periodic retraining.
Open, transparent, inclusive	Open weights and MIT code; each suggestion shows its contributing signals and similar images; seed data spans many regions; human approval required.

4. Planned testing methodology

- **Data protocol:** Commons files from ~100 seed categories (objects, scenes, events, abstract concepts, and regions across Asia, Africa and Latin America) plus subcategories. Split by upload time (≈70/15/15); remove test images that are perceptual-hash or embedding (cos ≥ 0.95) near-duplicates of training images; optional uploader-disjoint split.
- **Primary metrics:** sample-wise P@k, R@k and F1@k for $k \in \{1, 3, 5, 10\}$; macro mAP over categories; reported both *with* metadata and *image only*.
- **Tail and breadth:** recall@k grouped by training frequency (unseen / tail / torso / head) and by category type (object / scene / event / abstract); label coverage of the vocabulary as an upper bound.

- **Ablations:** each scorer alone vs. fusion, with and without hierarchy post-processing, to quantify what each component adds.
- **Hierarchy-aware precision:** a predicted ancestor or descendant of a true label counts as partially correct; a manual review of a sample of “wrong” suggestions estimates how many are correct-but-unlabelled.
- **Robustness:** metrics stratified by original resolution, and under synthetic Gaussian blur, heavy JPEG compression and extreme downscaling.
- **Efficiency:** p50/p95 end-to-end latency per image, scoring throughput, index memory; comparison of CLIP ViT-B/32 vs. SigLIP encoders.
- **Continual learning check:** add new categories and accepted feedback at run time and verify they are suggested without retraining.
- **Engineering tests:** automated unit tests (category filtering, metrics, index search, hierarchy rules) and an offline end-to-end test that runs on every change.